Predicting NBA Player Points Using Machine Learning Techniques

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Link:

haring

Introduction



Original Research Questions

- Can we predict an NBA player's total points for the upcoming season based on historical performance data?
- Which performance metrics are most indicative of future point-scoring ability?

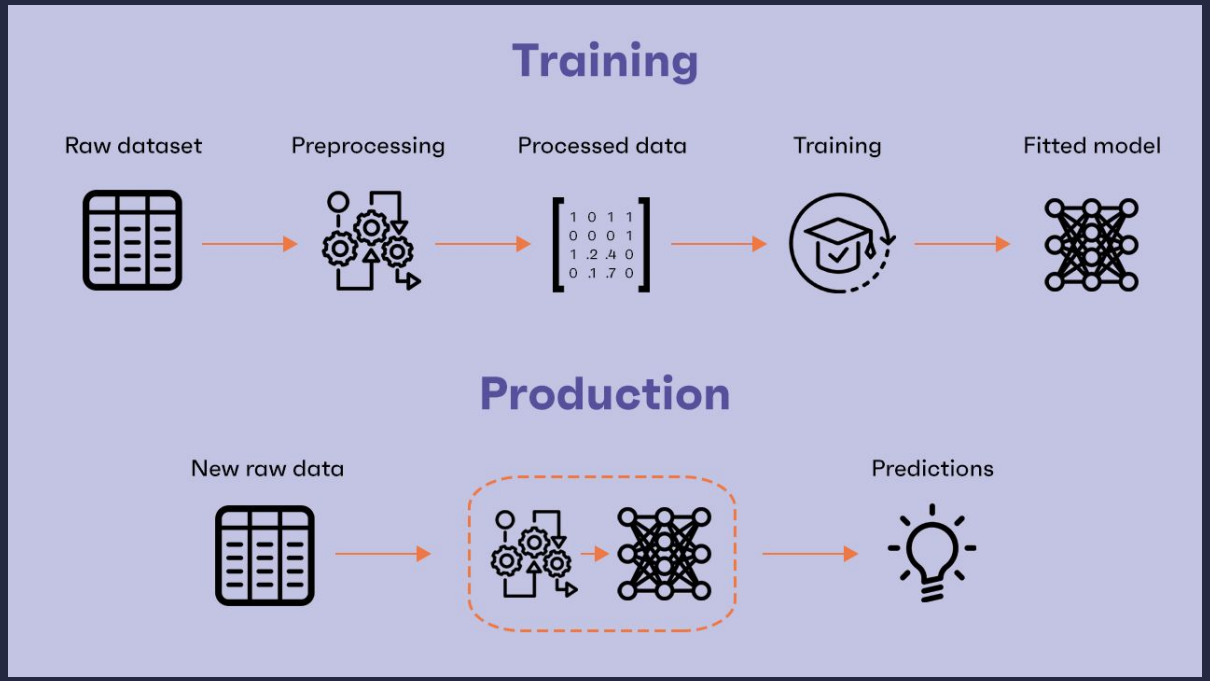
Final Research Question

What are the key contributing factors that could predict an NBA player's future scoring and overall on-court performance with a high degree of accuracy?

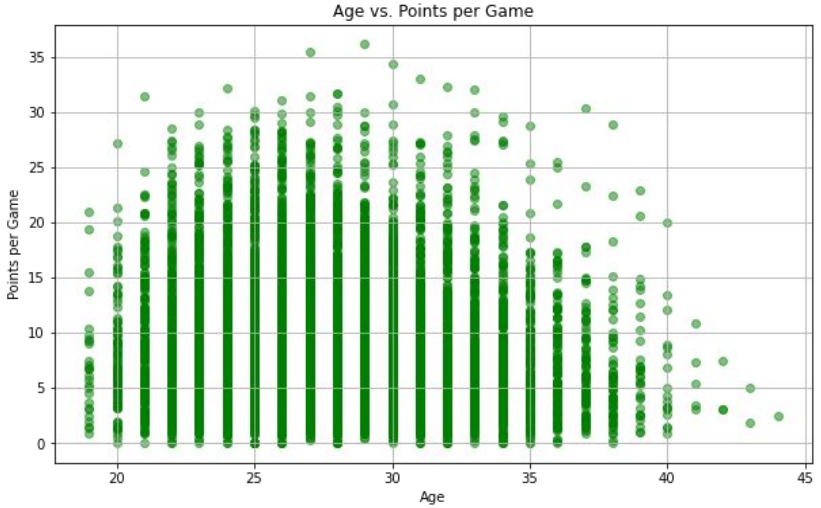
Data Sources

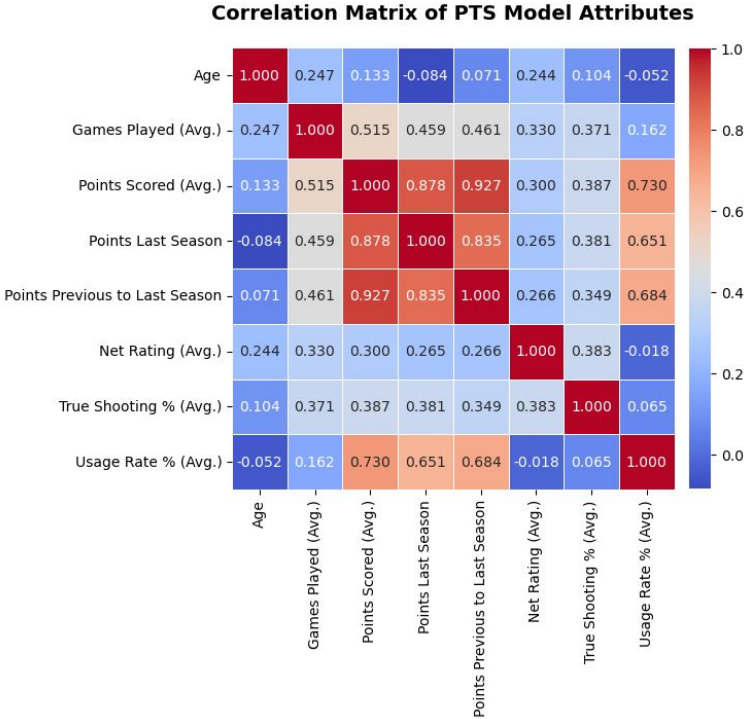


Data Pre-Processing

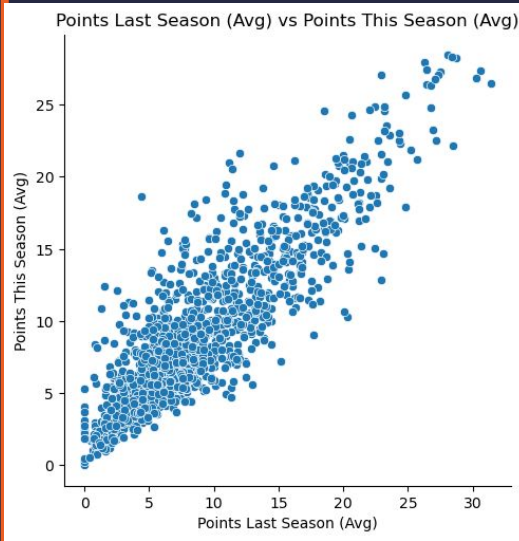


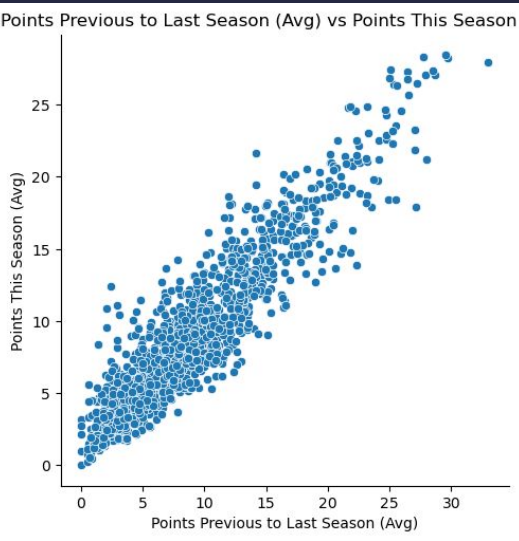
Preliminary Results

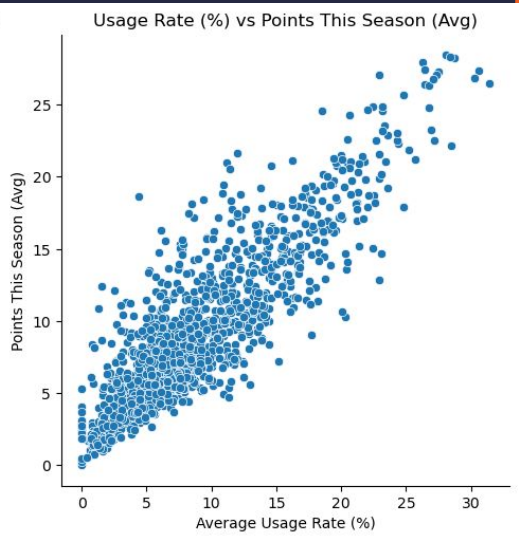




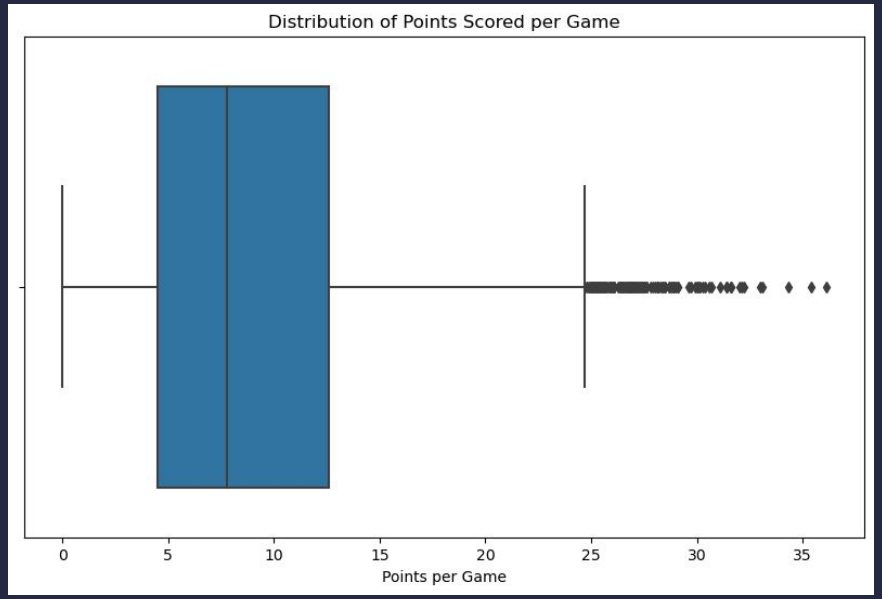
Results


Points Last Season (Avg) vs Points This Season (Avg)
Points This Season (Avg)
Points Last Season (Avg)
0
5
10
15
20
25
30


Points Previous to Last Season (Avg) vs Points This Season (Avg)
Points This Season (Avg)
Points Previous to Last Season (Avg)
0
5
10
15
20
25
30


Usage Rate (%) vs Points This Season (Avg)
Points This Season (Avg)
Average Usage Rate (%)
0
5
10
15
20
25
30

Results Cont'd



- MSE: 14.54
- RMSE: 3.81
- R^2: 0.62

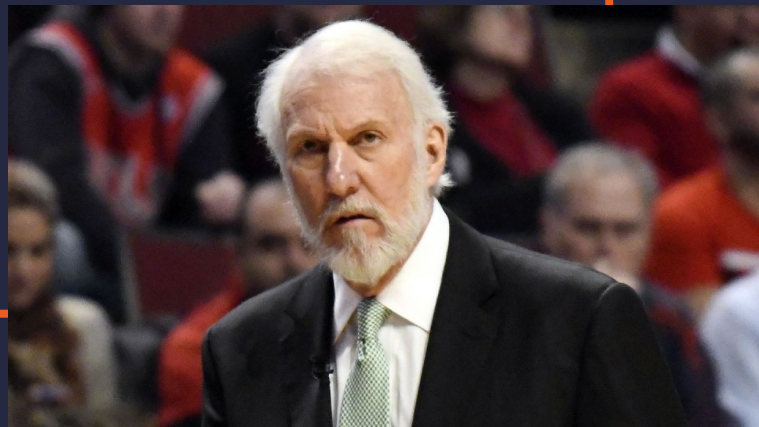


Limitations

1. Limited feature set = Limited Player Characterization.

2. Non-player confounding variables (e.g. coach, team dynamic, etc.)

3. Correlation does not imply causation.



Future Work

- Update on a game by game basis
- Predict for other sports?



Conclusion